

10.4.5.1. PreferredExtendedPanID Attribute

This attribute SHALL represent the Thread Extended PAN ID value for the Thread network designated by the user to be their preferred network for commissioning of Thread devices. If not null, the value of this attribute SHALL match the ExtendedPanID of a network in the ThreadNetworks attribute. A write operation with a non-null value that does not match any network in the ThreadNetworks list SHALL be rejected with a status of CONSTRAINT_ERROR.

The purpose of designating one Thread network as preferred is to help a commissioner to select a Thread network when a Thread device is within suitable range of more than one Thread network which appears in the ThreadNetworks list. A value of null indicates that there is no current preferred network: All networks MAY be treated as equally preferred by a commissioner with access to this cluster.

This attribute MAY be automatically set to the ExtendedPanID of the first Thread network added to the ThreadNetworks list.

A client SHALL obtain user consent before changing the value of this attribute from a non-null value.

On a factory reset this attribute SHALL be reset to null.

10.4.5.2. ThreadNetworks Attribute

This attribute SHALL indicate the list of Thread Networks known about by this cluster. If the node hosting this cluster includes a Thread Border Router, then an entry for its Thread Network SHALL be included in this list.

The list can be modified via the [AddNetwork](#) and [RemoveNetwork](#) commands.

For each entry in the list, the cluster server also stores a Thread Operational Dataset. Clients use the [GetOperationalDataset](#) command to obtain the Operational Dataset for an entry in this list.

On a factory reset this list SHALL be cleared, and any Thread Operational datasets previously stored SHALL be removed from the Node.

10.4.5.3. ThreadNetworkTableSize Attribute

This attribute SHALL indicate the maximum number of entries that can be held in the ThreadNetworks list; it SHALL be at least 2 times the number of SupportedFabrics advertised in the Operational Credentials Cluster on the root endpoint of this node.

10.4.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	AddNetwork	client ⇒ server	Y	M T	M
0x01	RemoveNetwork	client ⇒ server	Y	M T	M